



Platform Strategy, Differentiation & System Blueprint

EduHackers 2.0 consolidated concept + market research
September 2026

Purpose

Define what Vossie is, how the two applications work together, where it differs from existing marketplaces, and what must be validated before implementation.

Strategy document - not a claim of production readiness

1. Executive summary

Vossie has evolved from a campus buy-and-sell concept into a closed student commerce, logistics and entrepreneurship platform. The core architecture is one platform with two application families sharing identity, permissions, orders, payments, custody, realtime events, analytics and audit infrastructure.

Core thesis: Vossie's differentiation is not merely 'a marketplace for students'. The stronger proposition is verified campus identity + multi-role student participation + campus logistics/custody + cross-campus transfers + operational tooling + demand intelligence for student entrepreneurship.

The student-facing Vossie Marketplace combines Buyer, Seller and Courier modes under one verified student identity. Vossie Operations serves hub staff, transfer operations, Incubation, support, moderation, finance, compliance, security and administrators through separate permission-controlled workspaces.

The existing engineering package is extensive but is not a finished application: it specifies 90 database tables, 202 API operations, 177 screens, 514 action bindings and 385 delivery tickets. A restricted backend foundation exists, while most screens, handlers, provider integrations and production deployment remain implementation work.

2. Product architecture

Vossie Marketplace - student application

One identity, progressive roles. A student should not create three unrelated accounts. After student verification, Buyer access can be enabled. Seller and Courier capabilities are activated through additional role-specific verification, agreements and payout/operational setup.

Mode	Primary responsibility	Key experiences
Buyer	Discover and purchase within the verified campus network.	Explore, search, product/service pages, chat, checkout, orders, tracking, collection, disputes, ratings.
Seller	Operate a student storefront/business.	Listings, inventory, requests, orders, fulfilment, earnings, payouts, customers and business analytics.
Courier	Perform authorised campus/local delivery work.	Offers, acceptance, pickup, route, QR/PIN handover, proof, incidents and earnings.

Vossie Operations - institutional and build-team application

Operations should be one staff application with granular grants, not a collection of disconnected admin apps. A user's workspace is determined by authorised memberships and resource scope.

Hub: intake, package scans, condition checks, custody, pickup/release, outgoing transfers, returns and exceptions.

Transfers: manifests, hub-to-hub routes, package scans and mismatch handling. **Incubation:** seller/business health, campus demand, unmet searches, trends, interventions and growth plans. **Operations/Admin:** command centre, users, sellers, couriers, listings, orders, payments, disputes, tickets, compliance, incidents, audit, security, configuration and platform health.

3. Identity, authentication and security

The preferred long-term model is institutional federation rather than Vossie reproducing the institution's authentication stack. Where Eduvos identity is backed by Microsoft/Entra, the intended flow is Vossie -> institutional sign-in -> institutional MFA/Authenticator -> verified identity result -> Vossie session. Vossie should never receive or store the student's Authenticator secret.

For a zero-budget prototype, this must be an explicitly labelled sandbox simulation: student number / institutional identity -> simulated Authenticator approval -> verified student state -> campus association -> Vossie account. Student-card information can support verification or campus interactions, but a photographed card should not become the primary long-term authentication mechanism.

Production authorisation must be enforced server-side. A hidden UI button is not access control. Sensitive operations - payout destination changes, staff grants, high-risk refunds, MFA changes and certain financial overrides - should use step-up authentication, strict audit records and, where appropriate, two-person approval.

4. Payments and transaction security

Vossie should orchestrate commerce without storing raw card numbers, CVVs or authentication secrets. A regulated payment provider should handle payment credentials. Vossie stores transaction references, status, amounts, fees, allocations, refunds and payout state.

The money model should use auditable ledger journals/lines rather than repeatedly mutating a single balance. Payment callbacks must be idempotent, and order state, money state and custody state must remain distinct.

Prototype payment journey: Checkout -> provider/sandbox authorisation -> Payment secured -> Seller preparing -> Fulfilment/custody -> Buyer receipt confirmation -> Transaction completed -> Seller/courier payout eligibility. Refund and dispute paths must also be demonstrable.

5. Cross-campus commerce and custody

Cross-campus operation is one of the strongest differences between Vossie and a conventional classifieds experience. A buyer at one campus can discover an item at another campus and choose an authorised fulfilment path: collection, eligible courier delivery, or hub-to-hub transfer.

A robust transfer path records Seller -> Origin Hub -> QR custody event -> Manifest/Transfer -> Destination Hub -> QR custody event -> Buyer -> recipient QR/PIN confirmation. Each handoff produces evidence for tracking, exceptions and disputes. Offline clients must not falsely confirm authoritative financial or custody events.

6. Incubation intelligence

Incubation is not the seller's ordinary dashboard. Seller Mode provides the seller's own listings, orders, inventory, earnings and analytics. Incubation receives appropriately scoped aggregate intelligence about demand, unmet searches, stock-outs, category trends, business health and intervention opportunities.

This creates a feedback loop unavailable in a simple listings app: marketplace behaviour can reveal unmet campus demand; the Incubation team can use those signals to support student businesses; new supply can then return to the marketplace.

7. Competitive landscape - September 2026 research

The comparison below focuses on currently visible marketplace capabilities, not brand size or overall quality. Competitors solve parts of the same problem, but Vossie's proposed system combines several layers that are usually separate.

Platform	What it currently demonstrates	Where Vossie's proposed model differs
Yaga (SA)	Integrated marketplace payments, buyer/seller protection, wallet-style seller proceeds and multiple national courier options. Payment is held until receipt confirmation.	Vossie is institution-scoped and adds verified student roles, student couriers, campus hubs, custody events, hub-to-hub transfer and Incubation intelligence. Yaga is primarily a general fashion marketplace.
Bob Shop (SA)	Large general marketplace with seller ratings, in-platform payments, buyer protection and delivery/transit mechanisms.	Vossie narrows trust to an institutional community and integrates campus identity, campus operations, student delivery roles and institution-facing operational intelligence.
Campus Marketplace	Student-only marketplace positioning with campus restriction, listings, chat, search and protected Paystack payments.	This is the closest conceptual category. Vossie's proposed moat must therefore come from operational depth: progressive multi-role identity, hubs/custody, cross-campus transfers, courier workflow, staff Operations and Incubation.
UniBay (SA)	South African student marketplace with listings, chat and study-note discovery/upload.	Vossie extends beyond discovery and messaging into transactional fulfilment, verified custody/logistics, seller operations and institution-side workflows.
Student Hub Marketing	Campus-focused buy/sell experience with authentication, profiles, listings, chat and campus-restricted discovery.	Vossie's differentiation again depends on the system behind the listings: transaction state, payments/ledger, courier assignment, hubs, transfers, evidence, disputes and entrepreneurship intelligence.

What the research changes

Student-only access by itself is **not unique**. Current apps already market verified/campus-restricted student marketplaces. Likewise, chat, listings, search, ratings, protected payments and courier integrations are established marketplace features. Vossie should not present those as its primary moat.

The more defensible differentiation is the **combination** of: institutional identity and progressive role activation; Buyer/Seller/Courier participation under one identity; campus hubs as operational nodes; custody-aware QR/PIN handoffs; cross-campus manifests and transfers; an institution-facing Operations console; and Incubation intelligence generated from marketplace demand.

Positioning recommendation: do not pitch Vossie as 'Facebook Marketplace for students'. Pitch it as a verified campus commerce network that connects student demand, student businesses, student delivery, campus logistics and entrepreneurship support in one operational system.

8. Differentiation framework

Layer	Vossie proposition	Strategic value
Trust	Institution-linked identity, student verification, progressive Seller/Courier verification and role-scoped permissions.	Reduces anonymous marketplace interactions and creates a defined community boundary.
Commerce	Listings, requests, protected checkout, transaction states, refunds, payouts and ledger.	Moves beyond informal DM-based trading.
Logistics	Student courier + campus hub + hub-to-hub transfer + QR/PIN custody.	Turns campuses into fulfilment nodes and creates evidence at handoffs.
Operations	Tickets, disputes, moderation, finance, compliance, security, audit and platform monitoring.	Makes the marketplace institution-operable rather than merely student-facing.

Layer	Vossie proposition	Strategic value
Entrepreneurship	Demand signals, unmet searches, seller health and Incubation interventions.	Converts marketplace data into student-business development signals.
Network	My Campus + cross-campus discovery and fulfilment.	Expands liquidity beyond one campus while preserving campus-aware logistics.

9. Visual and interaction direction

The existing Vossie visual system uses warm ivory #FAF8F2, navy #29475B, powder blue #AFCBDE, slate #567487 and pale sage #DDE5D9. The intended visual literacy is soft and restrained, while interaction patterns borrow the clarity and immediacy of on-demand apps: strong cards, obvious states, one primary task per screen, live status and recovery paths.

The design must remain accessible: persistent labels, visible focus, practical touch targets, reduced-motion alternatives, clear loading/empty/error/offline states and explicit confirmation for irreversible actions.

10. Current implementation reality

Specified: domain/data contracts, screen/action mappings, API inventory, transaction rules, role boundaries, testing expectations and implementation backlog.

Implemented foundation: SQL migrations/invariants and a restricted TypeScript/NestJS ordinary-product flow covering parts of quote/direct request, seller acceptance, capture-to-ledger processing, preparation, package creation and hub receipt.

Still outstanding: most API handlers, complete native/web UI, ranked matching, full provider checkout/refunds/payouts, media pipeline, complete courier and extension workflows, production hosting and external integrations.

The existing demo therefore remains a reviewable prototype. Identity, payment, camera, location and messaging simulations do not prove that production services are connected.

11. Decisions to resolve in the next work session

1. Pilot campus(es) and which physical hubs are in scope.
2. Exact institutional identity/SSO path and what Eduvos can officially expose.
3. Buyer, Seller and Courier registration requirements and verification evidence.
4. Payment provider, merchant model, settlement/custody rules, fees, refunds and payout timing.
5. Which fulfilment methods launch first: direct collection, student courier, hub pickup, cross-campus transfer.
6. Courier eligibility, safety controls, assignment rules and incident process.
7. Hub operating procedures, custody evidence, storage limits and exception handling.
8. Marketplace categories permitted at launch and prohibited/restricted listing policy.
9. Incubation metrics that are useful without exposing private/internal scores to sellers.
10. Operations permission matrix, two-person approvals, audit retention and support ownership.
11. Zero-budget prototype stack versus costs that only begin at pilot/production scale.
12. MVP boundary: which subset of the 177-screen architecture proves the end-to-end system first.

12. Recommended proof-of-flow

Before attempting every screen, the next prototype should prove one complete ordinary-product journey with real state logic: verified student sign-in simulation -> Buyer discovery -> Seller request/acceptance -> sandbox payment -> package creation -> hub intake -> courier/transfer handoff -> destination/recipient confirmation -> ledger/payout state -> dispute/evidence path -> Operations audit trail.

Then deliberately test failure and concurrency cases: two sellers accepting the same request, two couriers taking one job, duplicate payment callbacks, expired checkout holds, wrong QR scans, repeated recipient PINs, offline custody attempts and transfer mismatches. The quality of these state transitions matters more than adding another polished dashboard.

13. Research notes and sources

Market research checked 27 September 2026. Current competitor facts were verified against available official/help or app-store material. Product capabilities can change, so this section should be refreshed before an external pitch.

- Yaga South Africa Help Center - How Yaga works; Buyer Protection; Payment Protection; Delivery; Fees Guide (2026 updates).
- Bob Shop - Buyer Protection Programme and related support documentation.
- Apple App Store South Africa - Campus Marketplace, UniBay, Student Hub Marketing listings.
- Internal Vossie / EduHackers materials - Vossie Developer Architecture and Build Guide; Vossie Full UI/UX Page Architecture; Vossie Implementation Specification; EduHackers Developer Build Workflow Technical Research Dossier.

14. Pilot Policy Defaults & Money Model

Working decisions from the pre-build research session. These are product rules to validate during the pilot, not final financial or regulatory assumptions.

14.1 Revenue model selected

Vossie uses transaction-based percentage revenue. A percentage is deducted from the seller's transaction earnings, and a percentage is deducted from the courier's delivery earnings. Under the current working model, the buyer is not charged a separate Vossie percentage fee.

The exact seller and courier percentages remain undecided. They should be set only after payment-provider costs, tax/accounting treatment, courier economics and pilot affordability are researched.

14.2 Seller earnings workflow

Event	Seller view	Meaning
Seller accepts request/order	Pending earnings	Expected proceeds are visible, but no payment has been confirmed.
Buyer payment confirmed	Secured	Provider confirms payment. Seller allocation is recorded but is not yet available for payout.
Order fulfilled / receipt confirmed	Available for payout	Fulfilment condition has been satisfied and no blocking dispute applies.
Payout processed	Paid out	Provider/payout rail has processed the seller's eligible allocation.

Critical distinction: seller acceptance can create **expected/pending earnings**, but the UI must not call money 'secured' until the payment provider has actually confirmed the buyer's payment.

14.3 Courier earnings workflow

Courier earnings mirror the seller model: **Pending** when a delivery is accepted, **Secured** when the delivery allocation is funded, **Available** after the required delivery confirmation, and **Paid out** after payout processing. Vossie's courier commission is recorded separately as platform revenue.

14.4 Ledger view

The product should distinguish **Pending, Secured, Available and Paid Out** instead of displaying one ambiguous wallet balance. Internally, order state, payment state, custody state and payout state remain separate. Vossie's platform revenue, seller allocation, courier allocation, refunds and reversals should be represented as auditable ledger entries.

14.5 Workflow

1. REQUEST / ORDER	2. SELLER ACCEPTS	3. BUYER PAYS	4. FULFILMENT	5. RELEASE
Price and fulfilment terms established.	Expected seller earnings appear as Pending .	Provider confirms funds. Allocation becomes Secured .	Hub/courier/direct handoff generates custody or delivery evidence.	Receipt rule satisfied -> earnings become Available -> payout.

14.6 Next policy decision

The next unresolved rule is **what constitutes buyer receipt and therefore unlocks payout eligibility**: a manual 'Order received / Everything is okay' confirmation, a successful delivery QR/PIN event, or a hybrid model where QR/PIN proves delivery and the buyer receives a short problem-reporting window. This requires research because it affects payout speed, disputes, fraud exposure and provider design.

This section will be kept concise and updated as the research session continues; it is intended to capture decisions and workflows rather than become another long-form specification.

15. Confirmed Pilot Money & Custody Workflow

The research session refined the pilot money model into the following working policy. Percentages are pilot hypotheses and remain subject to provider economics, legal/accounting review and pilot evidence.

Rule	Working Vossie policy
Seller commission	7% of seller transaction value; no listing fee or seller subscription in the pilot.
Courier commission	5% of courier delivery earnings.
Buyer fee	No separate Vossie percentage commission. Buyer pays the applicable delivery charge by default.
Earnings states	Pending -> Secured -> Available -> Payout Processing -> Paid Out.
Payout choice	Seller/courier chooses Automatic or Manual withdrawal. Only Available funds are eligible.
Delivery proof	Package QR records custody; recipient OTP proves final delivery.
Inspection	OTP delivery starts a 24-hour Vossie Purchase Protection issue window . OTP proves delivery, not product satisfaction.
Dispute	Issue opened in time freezes affected seller funds; structured evidence + seller-first resolution; unresolved cases escalate to Operations.
Returns	Responsibility-based: seller fault -> seller pays; buyer fault/change-of-mind where allowed -> buyer pays; hub/courier fault -> logistics incident; unclear -> Operations adjudicates.
Full refund	Seller commission reverses on a full approved refund; processor costs remain separately accounted for.

Custody and release workflow

Seller -> Hub [package QR] -> Courier [package QR] -> Buyer [delivery OTP] -> Delivered -> 24-hour issue window -> Completed -> Seller funds Available. A timely qualifying dispute pauses the affected seller allocation. Product complaints do not automatically freeze courier earnings when OTP-confirmed delivery itself was successfully completed.

16. Authentication Research & Prototype Workflow

Core security principle: an identifier format is not proof of identity. An EDUV-style student number or an address ending in an institutional domain can be useful validation data, but access must not be granted merely because a user can type a correctly formatted value.

Production target

The target architecture is institutional federation: **Vossie -> Continue with Eduvos -> Eduvos/Microsoft identity provider -> existing Microsoft Authenticator/MFA -> signed authentication response -> Vossie validates trusted issuer/tenant/token -> active-student eligibility check -> Vossie session.** Eduvos/Microsoft remains responsible for the student's institutional password and MFA. Vossie does not store or receive the student's password, Authenticator seed or approval code.

Authentication, eligibility and authorization remain separate. Authentication establishes control of the institutional account. Eligibility establishes that the person is currently permitted to participate as an Eduvos student. Authorization determines Buyer/Seller/Courier capabilities inside Vossie.

One identity, progressive roles

There are no separate Buyer, Seller and Courier login systems. A verified Vossie student identity receives Buyer access as the base marketplace capability. Seller and Courier are activated after login through additional role-specific onboarding, agreements, payout information and approval. One student can therefore hold Buyer + Seller + Courier capabilities simultaneously.

Prototype/demo workflow

Step	Demo experience	Production replacement
1	Student taps Continue with Eduvos .	Same UI entry point.
2	Prototype presents a clearly labelled sandbox institutional sign-in handoff.	Redirect to approved Eduvos/Microsoft institutional login.
3	Demo simulates the existing Microsoft Authenticator approval state; Vossie does not invent a second MFA system.	Microsoft/Eduvos performs real MFA/Authenticator challenge.
4	Sandbox identity adapter returns a test verified identity and active-student state.	Vossie validates signed token/tenant and approved student-status source.
5	New verified identity -> create Vossie profile. Existing identity -> open existing account.	Same application logic.
6	Buyer active by default; Seller/Courier activation available from profile.	Same role architecture with production verification.

Prototype boundary

The prototype must visibly label institutional authentication as **Sandbox/Demo** until Eduvos authorizes a real integration. It must not claim to verify live Eduvos credentials, call an unapproved student-information API, or collect real Eduvos passwords/MFA secrets. The sandbox should preserve the same interface contract so the mock identity adapter can later be replaced by the approved institutional integration without redesigning account/role logic.

No Vossie-specific Face ID or passkey requirement is included in the current design. Sensitive payout/account changes still require an appropriate re-authentication/step-up policy once the real institutional integration and payment provider are known.

Eduvos integration requirements to validate

Before production: confirm the institutional identity provider/tenant; obtain approval for the Vossie application; determine the minimum identity claims Eduvos will release; determine the authoritative method for active-student and campus status; define deactivation/revalidation behaviour when student status changes; agree privacy/data-retention requirements; and test failure/revocation flows with Eduvos IT/security.

17. Cross-Campus Hybrid Fulfilment — Confirmed Direction

Cross-campus fulfilment is hybrid by design. Vossie should not dispatch a dedicated campus-to-campus movement simply because one parcel exists. The system pools eligible parcels by route while preserving buyer choice when consolidation is inefficient or too slow.

Fulfilment choices

Method	When used	Custody / system treatment
Vossie campus transfer	Consolidated route has viable volume, capacity, economics and ETA.	Origin Hub QR -> locked manifest -> transfer -> Destination Hub QR/reconciliation.
External point / locker	Low-volume route or buyer prefers a third-party collection point.	Vossie records provider, quote, waybill/tracking ID, handover and provider events.
External door courier	Buyer wants direct delivery and compatible provider/service is available.	Provider executes logistics; Vossie translates tracking into buyer-facing states.
Student transfer courier	Only when route, transport, eligibility, safety/insurance and operating rules permit.	Package/manifest custody remains QR-audited; no unnecessary buyer/seller private data.

Dispatch logic

Do not hard-code a minimum parcel count. Route dispatch should eventually evaluate package count, weight/volume, oldest waiting time, scheduled capacity, promised ETA, Vossie transfer cost and available external-courier price. The previously proposed three daily windows remain configurable research defaults, not production policy.

Confirmed cross-campus path: Buyer purchases -> Seller prepares -> Origin Hub inspection + QR -> route/fulfilment method selected -> Vossie manifest transfer OR external shipment -> destination reconciliation/collection point -> local delivery or pickup -> buyer OTP where Vossie controls final handover -> Delivered -> 24-hour Vossie Purchase Protection window.

18. Payment Gateway & Financial Security Workflow

The payment architecture separates marketplace entitlement from movement of regulated funds. Vossie maintains the authoritative internal ledger for allocations and eligibility; the payment service provider (PSP) remains authoritative for payment execution, refunds and payouts. Vossie must not present itself as independently holding escrow funds unless the eventual regulated commercial arrangement explicitly permits that structure.

Confirmed transaction workflow

Stage	System behaviour
1. Checkout	Buyer enters payment through PSP-hosted/tokenised payment infrastructure. Vossie does not store raw card number or CVV.
2. Provider confirmation	PSP confirms server-to-server. Vossie verifies webhook authenticity, order reference, amount/currency and processes the event idempotently.
3. Ledger	Append-only journal records buyer payment, seller allocation, courier/logistics allocation where applicable, Vossie commission and processor cost as distinct components.
4. Secured earnings	Seller entitlement becomes Secured , not yet payout-eligible. Client UI cannot declare payment success.
5. Delivery	QR custody and final OTP evidence drive fulfilment state. OTP proves delivery, not product satisfaction.
6. Protection	Delivered order enters the confirmed 24-hour Vossie Purchase Protection issue window. A qualifying dispute freezes the affected seller allocation.
7. Available	No qualifying issue / resolved release -> seller entitlement becomes Available .
8. Payout	Automatic or manual payout is requested through the PSP to a verified payout destination; provider result moves state through Payout Processing -> Paid Out.

Provider strategy

Prototype direction: use a PSP sandbox through a provider-neutral **PaymentProvider** adapter and simulate the full Vossie ledger/earnings state machine. Paystack is the current sandbox candidate; Stitch, Payfast and Peach remain production candidates/components. Production selection requires written validation that the provider arrangement supports Vossie's delayed payout eligibility, seller/courier payouts, refunds, reconciliation and South African regulatory requirements.

Security controls

Non-negotiable controls: backend-only secret keys; PSP tokenisation/hosted card capture; authenticated webhooks; idempotency keys and duplicate-event protection; immutable financial journals; server-side RBAC; payout-destination verification; re-authentication/step-up for payout changes; financial audit events; daily/provider reconciliation; dispute and refund traceability; and no client-controlled payment status.

Financial state vocabulary

Pending -> Secured -> Available -> Payout Processing -> Paid Out. Product price, seller commission, courier commission, original delivery fee, external courier/transfer cost, processor fee and refunds remain distinct ledger components. Refunds create reversing/adjusting entries rather than deleting or rewriting historical journals.

19. Two-App Experience, Roles & Responsive UI/UX — Confirmed

App 1 — Vossie Marketplace: mobile-first student experience for Buyer, Seller Mode and Courier Mode. **App 2 — Vossie Operations:** web/desktop-first operational application with Seller Studio, Campus Hub, Incubation and Platform Operations workspaces. Both remain fully responsive and share the same backend, identity, authorization model and design system.

Authorization model

Permissions are enforced server-side as **Role + Resource + Action + Context**, not by hiding interface controls. Examples include SELLER + LISTING:OWN + UPDATE; HUB_OPERATOR + PACKAGE:ASSIGNED_HUB + ACCEPT_CUSTODY; COURIER + DELIVERY:ASSIGNED + SCAN_PICKUP. High-risk financial or privilege actions may require step-up authentication and/or a second approver.

Workspace / Role	Primary scope	Explicit boundary
Buyer	Browse, purchase, pay, track, message, collect, dispute.	No seller private analytics, manifests or admin data.
Seller Mode	Daily requests, orders, quick listing actions, earnings, fulfilment.	Cannot self-approve listings or alter ledger/dispute truth.
Courier Mode	Eligible jobs, pickup/QR, route, OTP handover, earnings.	Only assignment-necessary data.
Seller Studio	Store setup, listings, inventory, pricing, availability/timetable, analytics.	Academic schedule is not assumed to equal availability.
Campus Hub	Intake, condition, QR/custody, manifests, transfers, returns, exceptions.	Custody evidence only; no payout/order rewriting.
Incubation	Demand intelligence, business health, growth plans/interventions.	Permissioned programme data; no unrestricted seller/customer access.
Platform Operations	Support, moderation, disputes, finance, logistics, security, system health.	Separated duties; break-glass elevation is exceptional and audited.

Responsive UI/UX

Marketplace uses mobile-first navigation and expands to sidebar/multi-panel layouts on web. Operations is desktop/web-first for management and analytics while mobile remains task-focused for scanning, alerts, approvals and quick actions. The same workflows and permissions persist across iOS, Android, macOS/Windows browsers; layout and information density adapt to the device.

Timetable workflow

Seller uploads PDF -> isolated file validation/processing -> schedule extraction -> seller verifies/corrects class blocks -> seller adds personal availability constraints -> structured availability saved -> matching/booking engine consumes confirmed availability. Source timetable is not retained longer than justified.

20. Dynamic Inventory & Catalogue Engine — Confirmed

Vossie uses one marketplace core with category-specific inventory behaviours. Categories are not merely filters: selecting a category determines attributes, availability rules, fulfilment, search filters and seller form fields. Buyer discovery uses progressive category cards/chips rather than exposing the internal taxonomy.

Category	Examples	Inventory behaviour
Food & Drinks	Meals, snacks, baked goods, drinks, meal prep	Quantity + modifiers/add-ons + prep time + availability.
Essentials	Stationery, toiletries, res supplies, accessories, personal care	Stock: AVAILABLE -> RESERVED -> SOLD; checkout expiry releases reservation.
Fashion & Lifestyle	Clothing, shoes, bags, accessories, jewellery, beauty	Stock by variants such as size/colour/condition.
Services	Tutoring, design, coding, photography, beauty, repairs, printing	Time inventory: AVAILABLE -> HELD -> BOOKED; uses seller availability.
Rentals	Textbooks, calculators, electronics, cameras, equipment	Reservable assets/date ranges; pickup/return/overdue/damage states.
Digital	Notes, study guides, templates, designs, code, permitted academic resources	Private digital entitlement after payment; no public file URL.
Tickets & Events	Future student/campus events	Taxonomy designed but feature-flagged pending fraud/refund/event rules.

Seller creation flow

+ Create Listing -> Category -> Subcategory -> dynamic category form -> preview -> submit -> automated checks -> moderation when required -> publish. The seller sees only relevant fields. Core listing data is shared; category extensions provide food modifiers, fashion variants, service slots, rental reservation rules or digital entitlements.

Digital academic resources

Student-created notes/study resources require a rights/ownership declaration and appropriate moderation. Institutional exam papers, lecturer materials, textbook scans or third-party work are not automatically publishable merely because a student possesses a copy. Digital files remain private and purchase creates an authenticated entitlement.

Search, Operations & Incubation

Structured attributes power category-aware search and filters. Operations can add, reorder, disable or moderate subcategories without an app-store release when an existing approved schema supports the change; schema evolution is versioned. Search/view/purchase/stock-gap signals feed Incubation demand intelligence and seller opportunity analysis.

Locked inventory principle

Food = quantity + modifiers + preparation. Essentials = stock. Fashion = stock + variants. Services = time slots. Rentals = reservable assets + date ranges. Digital = access entitlements. Identity, payments, seller profiles, messaging, search, moderation, reviews and analytics remain shared platform services.

21. UI/UX Wireframe & Workflow Reference

Low-fidelity reference sketches for the confirmed Vossie architecture. These are interaction blueprints, not final visual comps. Marketplace remains mobile-first; Operations remains desktop-first with task-focused mobile adaptations.

Marketplace — Buyer Home / Discovery

VOSSIE MARKETPLACE — MOBILE	VOSSIE	MARKETPLACE — WEB
Potchefstroom ■ Buyer ■	Home	Potchefstroom ■ Buyer ■
Search products, services, notes...	Search	Search Vossie...
Categories: Food · Essentials · Fashion	Orders	Food Essentials Fashion Services Rentals Digital
Services · Rentals · Digital	Messages	Popular near you Recommended
Popular near you →	Profile	[Product] [Product] [Product] [Product]
[Product card] [Product card]		Active order: Preparing → Hub → Delivery
Home Search Orders Chat Me		

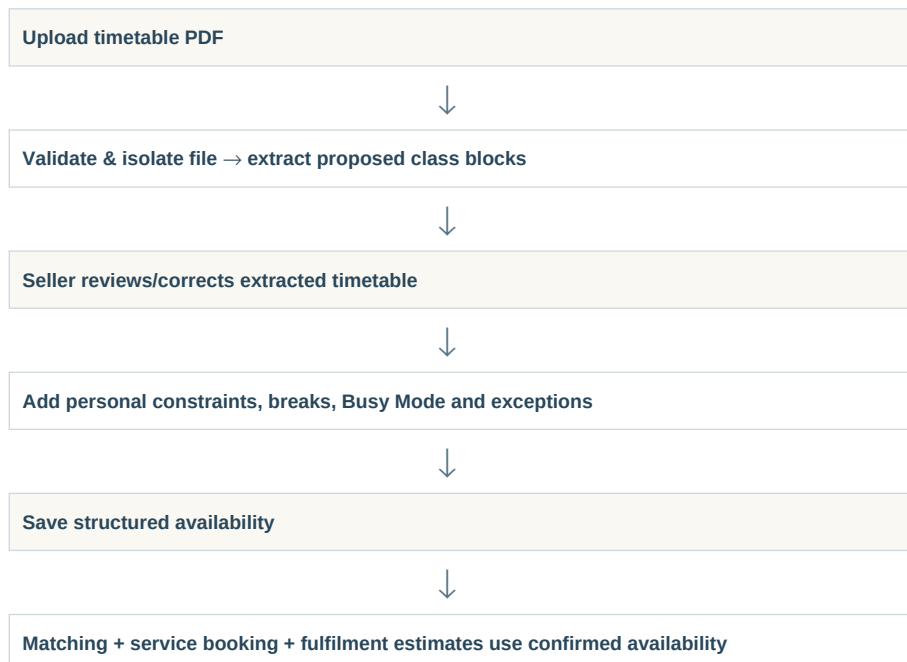
Marketplace — Seller & Courier quick operations

SELLER MODE	COURIER MODE
Today: AVAILABLE until 14:00	Available jobs: 4
New requests: 3	Potchefstroom Campus → Res
Orders requiring action: 2	R32 earning · 1 parcel
Listings: 14 live · 1 review	[ACCEPT JOB]
Earnings: Secured R465 · Available R920	Active: Scan pickup QR → Navigate → OTP
Requests Orders Listings Earnings	Jobs Active Scan Earnings

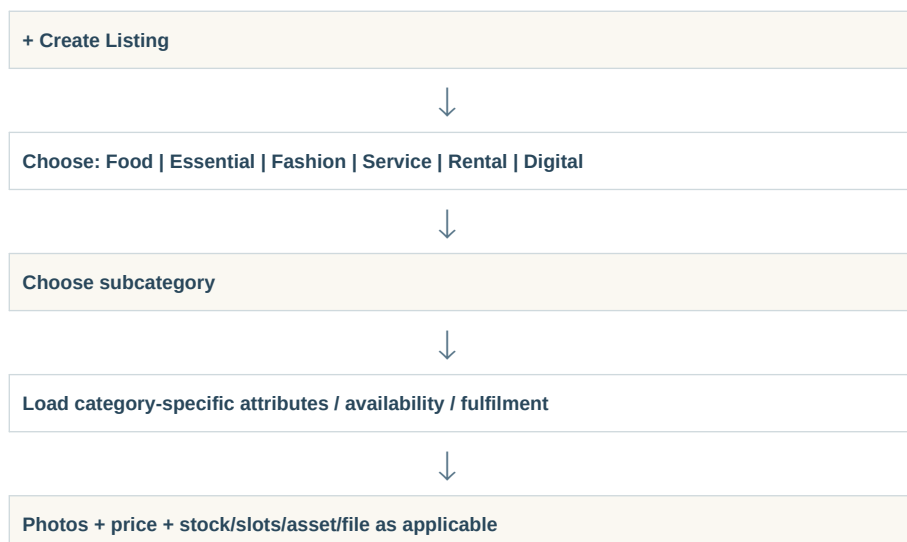
Seller Studio — Desktop Management

VOSSIE	SELLER STUDIO
Overview	Good morning + Create Listing
Store	Secured R2,840 Available R920 Orders 12 Actions 4
Listings	Today's availability: 08 10 12 14 16 18
Inventory	BUSY OPEN CLASS OPEN
Orders	Orders requiring attention
Availability	#VOS-1028 Ready for handover [OPEN]
Earnings	Demand insight: USB-C accessories searched 84× this week
Analytics	

Timetable → Seller Availability



Dynamic Listing Creation





Preview → Submit

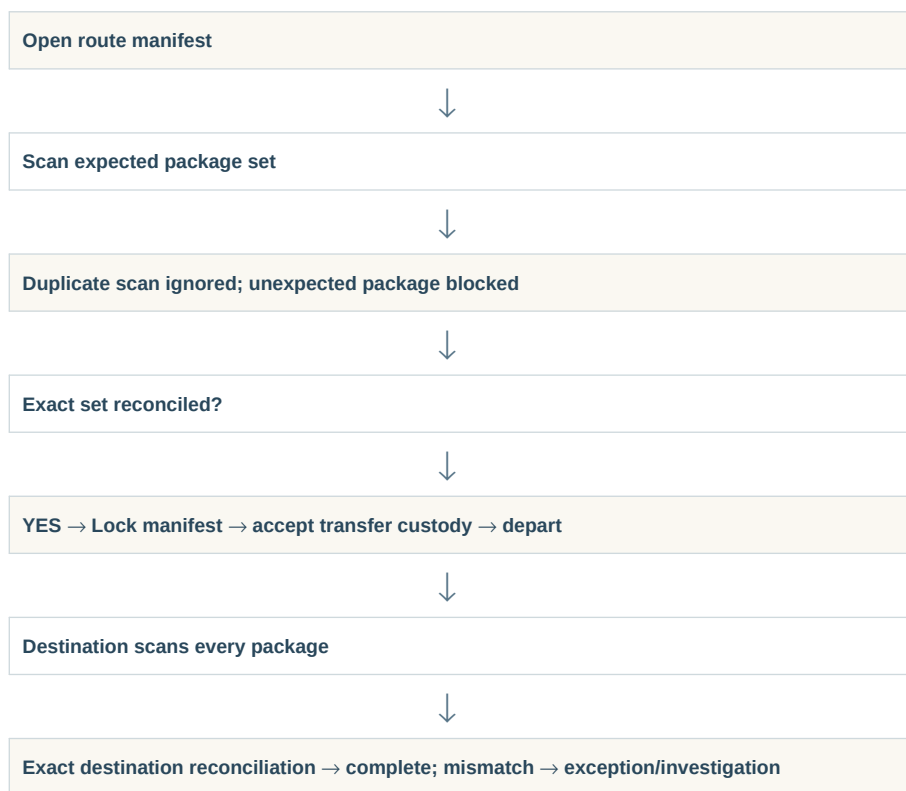


Automated checks → Publish OR Human Review

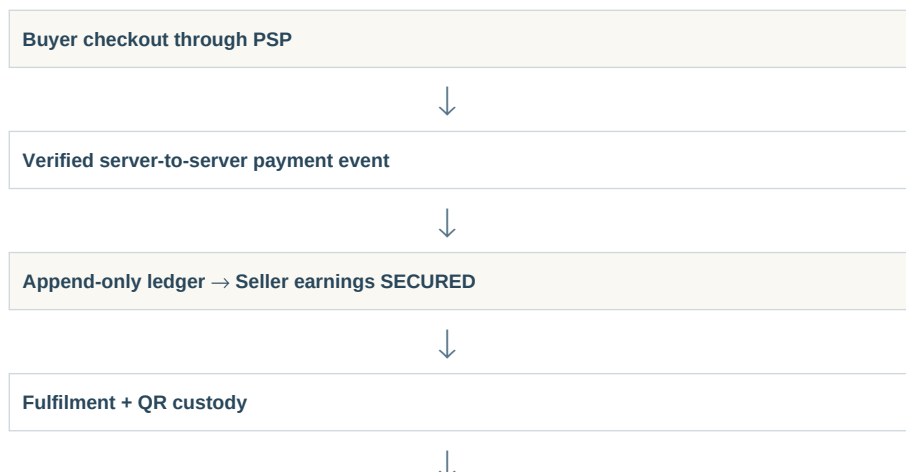
Campus Hub — Scan & Custody

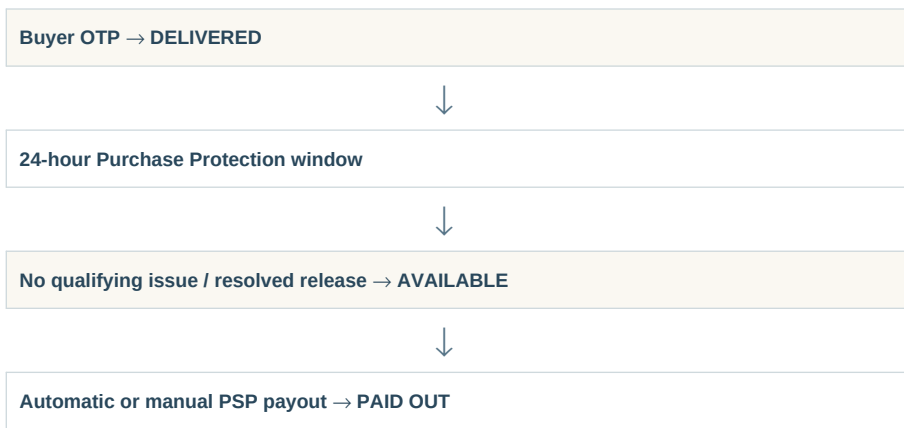
VOSSIE	CAMPUS HUB
Incoming	SCAN PACKAGE [CAMERA]
Ready	Package: VOS-PKG-20481 Expected: Seller Intake
Outgoing	Order verified ✓ Seller verified ✓
Transfers	Condition: ■ Intact ■ Damaged ■ Tampered ■ Exception
Returns	[REJECT + REASON] [ACCEPT CUSTODY]
Exceptions	On acceptance: activate/print authoritative package QR + custody event
SCAN	

Transfer Manifest & Reconciliation



Payment → Protection → Payout





Incubation — Business Intelligence

VOSSIE	INCUBATION
Portfolio	Campus: Potchefstroom ■ Period: 30 days ■
Business Health	Unmet demand: Printing ↑ USB-C Chargers ↑ Network Notes ↑
Demand	Participating sellers requiring attention: 6
Trends	Seller Growth Profile: demand · conversion · fulfilment · stock-outs
Unmet Searches	30-Day Plan: Goal → Action → Owner → Due Date → Result
Growth Plans	Aggregate intelligence by default; seller-level access permissioned
Interventions	

Platform Operations — Command Center

VOSSIE	COMMAND CENTER
Overview	Live Operations: Orders Transfers Open disputes Tickets
Users	Attention Required: payout exception · manifest mismatch · moderation
Sellers	Platform Health: API ✓ DB ✓ Queue ✓ Realtime ✓ PSP ✓
Couriers	Financial Ops: reconciliation · refunds · payout processing
Listings	Security: auth anomalies · role changes · webhook failures · incidents
Orders	
Hubs	
Transfers	
Payments	
Disputes	
Compliance	
Security	
Audit	
System	

Dispute / Ticket Investigation

VOSSIE	CASE #VOS-D-1042
Cases	Order #VOS-8821 Priority: High Assigned: Operator 14
Disputes	Buyer Seller Courier Hub Payment — linked case entities
Evidence	Timeline: Paid → Hub → Courier → OTP → Issue reported
Messages	Evidence: listing snapshot · QR events · OTP · photos · messages
Audit	Financial state: Seller allocation FROZEN
	[REQUEST INFO] [PARTIAL REFUND] [RETURN] [ESCALATE]

Security Operations / Vossie SOC View

VOSSIE	SECURITY OPERATIONS
Overview	Critical alerts: 2 Open incidents: 3 Failed webhook signatures: 4
Auth	AUTH_FAILURE spike PAYOUT_DESTINATION_CHANGED ROLE_CHANGED
Authorization	Authorization failures by endpoint / role
Financial	Custody anomalies: rejected QR · repeated OTP · manifest mismatch
Custody	No secrets/raw payment credentials displayed; structured events only
Incidents	
Audit	
System Health	

Responsive interaction rule

Desktop tables become mobile cards; sidebars become bottom navigation or drawers; multi-column investigations become staged detail views. Scanning and urgent operational actions remain one-tap on mobile. Permissions and workflow state do not change between devices.

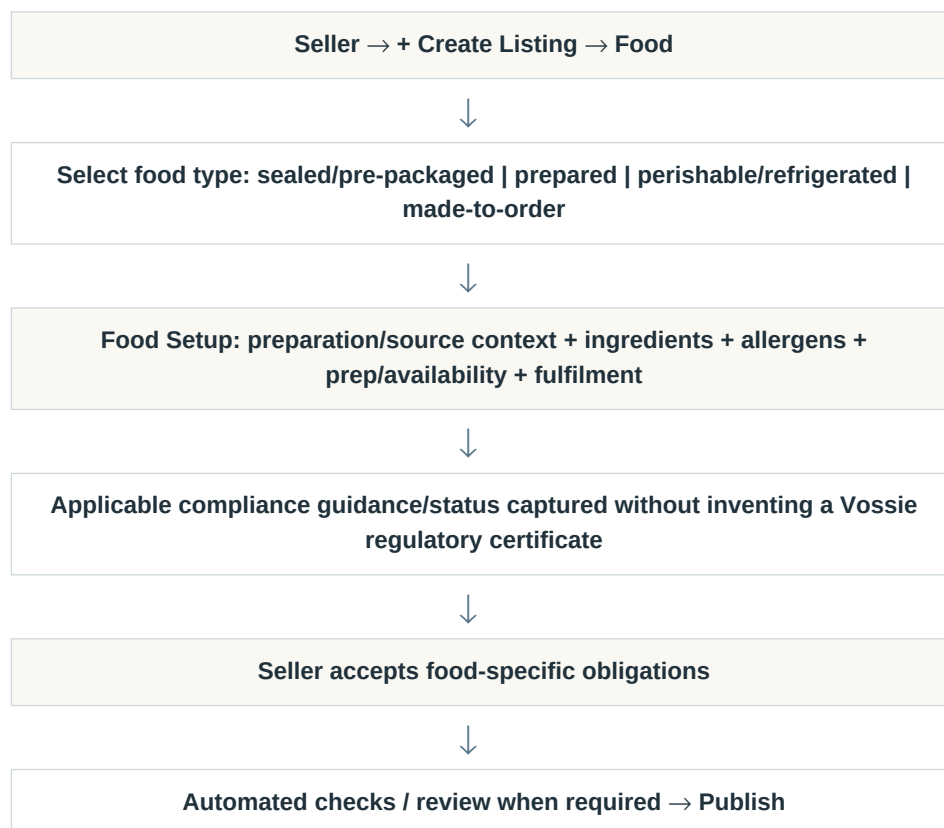
22. Food Marketplace Compliance, Trust & Safety — Confirmed Direction

Food remains available through the normal Seller capability; Vossie does not create a separate seller account merely because a student lists food. Food does, however, invoke a stronger category-specific workflow for applicable compliance, transparency, handling, traceability and incident response.

Product principle

Do not use an “eat/buy at your own risk” model. Buyer messaging should explain that food is supplied by independent student sellers while preserving buyer protection and reporting mechanisms. Seller terms allocate seller responsibilities, but the platform must not imply that consumer or food-law obligations have been waived.

Food listing onboarding



Compliance status

The platform may track **Food Setup Incomplete → Compliance Pending → Compliance Confirmed → Reverification Required**. “Confirmed” must only be shown when supported by the applicable evidence/process. Vossie should not encode a blanket student exemption. Before production launch, the operating model should be validated with Eduvos and the relevant Municipal Environmental Health Service.

Buyer-facing trust layer

Signal	Buyer experience
Identity	Student Verified; seller profile and order history.
Food transparency	Ingredients, known allergens, dietary claims where substantiated, prepared/availability information.
Reputation	Food-specific rating, completed food orders, order accuracy, packaging and readiness feedback.
Fulfilment	Pickup/local delivery, preparation estimate and timestamped handover/delivery.

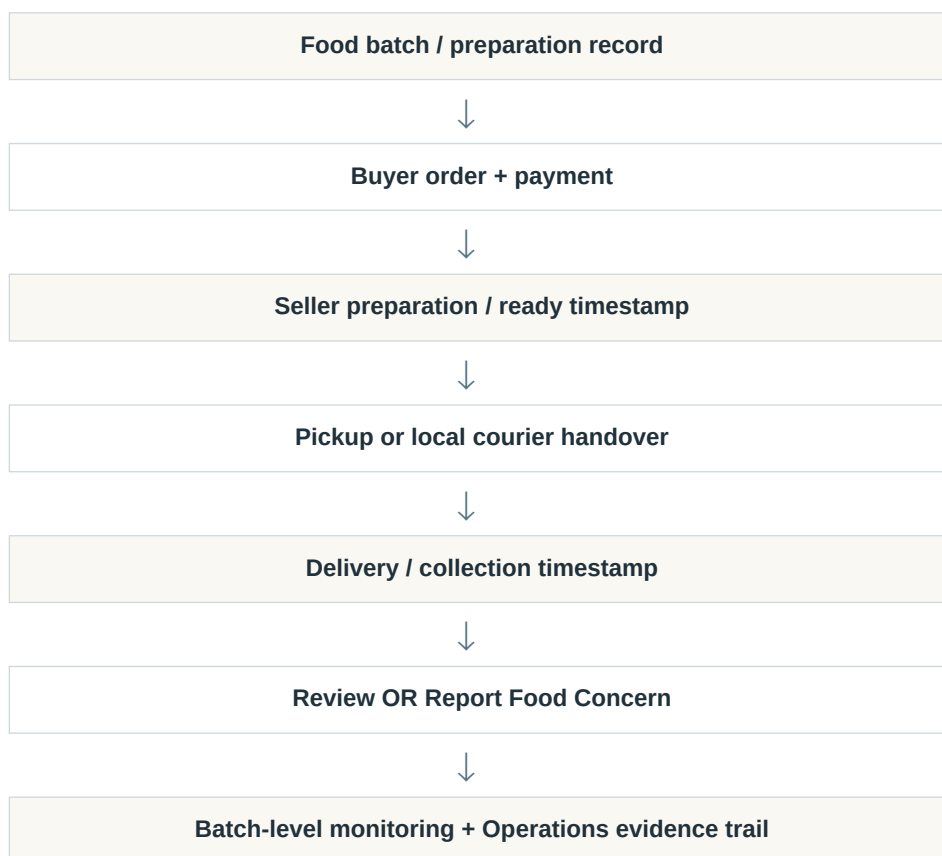
Signal	Buyer experience
Protection	Dedicated Report Food Concern route plus applicable Vossie purchase-protection process.

Seller obligations

Food-specific Seller Terms should require truthful product information; accurate ingredient/allergen disclosure to the extent applicable; compliance with applicable food laws; appropriate preparation, storage and transport; truthful dietary/safety claims; cooperation with incident investigations; and prompt disclosure of known contamination or safety concerns. These are marketplace obligations in addition to applicable law, not substitutes for it.

Prepared-food inventory & traceability

Prepared food should be batch/time-aware rather than persistent ordinary stock. A preparation batch records seller, listing, quantity, preparation/availability timestamps, affected orders and fulfilment events. Remaining prepared-food availability is removed at the appropriate cutoff instead of carrying stale inventory into the next day.



Food safety incident workflow





Fulfilment constraint

Prepared meals should begin as **same-campus/local pickup or local courier fulfilment**, not the ordinary inter-campus parcel-transfer network. A Campus Hub should not casually store prepared food alongside general parcels. Any future hub food handling requires its own approved operating model and facilities.

Operations workspace

Platform Operations → Compliance gains a Food view: **Food Listings, Compliance Status, Open Concerns, Batch Alerts, Seller Incidents, Affected Orders, Evidence, Escalations and Actions**. Actions are permissioned and audited.

Launch validation

Before a production Eduvos rollout, confirm the student-food operating model with Eduvos and the relevant municipal Environmental Health Service, including the practical route for applicable food-premises requirements. Final consumer-facing terms and allocation of responsibility should receive South African legal review.

Locked food principle

Light on the student-facing surface; controlled underneath. Any Seller can enter the Food workflow, but Food receives category-specific compliance guidance, seller obligations, structured disclosures, batch/time traceability, local fulfilment controls, buyer protection and rapid incident response. Vossie communicates evidence and traceability rather than making an unsupported promise that food is “safe.”

23. Inventory Taxonomy & Listing Presentation — Confirmed

The inventory taxonomy and shared listing presentation system are now locked. Buyer discovery stays simple while the catalogue remains structured underneath: **Category** → **Subcategory** → **Item/Service Type** → **Attributes** → **Variants/Modifiers** → **Availability** → **Fulfilment**.

Pilot catalogue

Category	Initial subcategories / types	Inventory behaviour
Food & Drinks	Meals; Fast Food; Snacks; Baked Goods; Desserts; Drinks; Meal Prep; Breakfast; Traditional; Healthy	Batch/time-aware quantity, modifiers, prep time, allergens, pickup/local delivery
Essentials	Stationery; Toiletries; Cleaning; Residence Supplies; Electronics Accessories; Personal Care; Groceries; Study Supplies	Physical stock: Available → Reserved → Sold
Fashion & Lifestyle	Clothing; Shoes; Bags; Accessories; Jewellery; Beauty; Hair; Sportswear	Variant stock by size/colour/condition
Services	Tutoring; Design; Coding & Tech; Photography/Video; Beauty/Hair; Repairs; Printing; CV/Career; Academic Support; Event Services	Time inventory: Available → Held → Booked
Rentals	Textbooks; Calculators; Electronics; Cameras; Tools; Event Equipment; Gaming; Sports Equipment; Formalwear	Date/time availability + deposit + condition + return state
Digital	Notes; Study Guides; Practice Resources; Templates; Design Assets; Code/Software; Digital Art; Productivity Resources	Payment-confirmed entitlement → My Library

Catalogue controls

Operations maintains the taxonomy through a versioned Catalogue/Taxonomy Manager: add, rename, reorder, disable, feature or moderate subcategories; assign approved listing schemas, filters, fulfilment modes and required attributes. Tickets & Events remains feature-flagged for later. Accommodation, Jobs, Cars and Financial Services are not ordinary inventory categories.

Shared buyer-facing listing grammar

Sellers provide structured data; they do not manually design marketplace cards. Vossie composes a consistent card, detail page and storefront. Core fields: cover image/gallery, title, seller, description, current price, promotion state, rating/reputation, availability and fulfilment.

BUYER CARD	DETAIL PAGE
[COVER IMAGE] ■	[HERO IMAGE] [gallery thumbnails]
Thato's Kitchen ✓ Chicken Alfredo	Chicken Alfredo Thato's Kitchen ✓ · ★ 4.8 · 138 sold
R55 R70 SPECIAL · SAVE R15	R55 R70 · 21% OFF · SAVE R15
★ 4.8 (42) · 20–30 min	Description · Size · Add-ons · Ingredients/Allergens
	Ready 20–30 min · Qty – 1 + ADD TO CART · calculated total

Category-specific transaction blocks

Type	Buyer controls
Food	Size/portion, modifiers/add-ons, ingredients/allergens, prep time, today's availability, pickup/local delivery.
Physical / Fashion	Variant chips/swatches, size, colour, condition, stock, delivery/pickup.
Service	Duration, price basis, online/in-person, available slots, Book Session.
Rental	Rental dates, daily/hourly rate, deposit, condition, collection/return, Check Availability.
Digital	File/resource type, preview, rights/moderation status, Buy Digital Resource → Open in My Library after entitlement.

Seller listing editor + live preview

CREATE LISTING	LIVE BUYER PREVIEW
Photos [+][+][+][+][+] Cover image: Image 1	[IMAGE]
Title [Chicken Alfredo] Category [Food & Drinks ■] Subcategory [Meals ■]	Chicken Alfredo
Description [...]	R55 R70 21% OFF · SAVE R15
Pricing Regular price [R70] ■ Run a special Special price [R55] Start [date/time] · End [date/time]	Seller sees buyer-facing result while editing.
Category fields appear dynamically below.	

Promotions

Promotions are structured fields, not text embedded in listing titles. Initial promotion types: **fixed price reduction**, **percentage discount**, **bundle/deal**. Seller enters regular price plus promotion terms/start/end; Vossie calculates and displays the saving and automatically restores the normal price when the promotion ends. Price history is retained internally to support promotion integrity.

Seller Studio listing management

Active | Draft | Under Review | Sold Out | Paused | Specials. Each row/card shows photo, title, current price, stock/availability, views, orders, revenue and status, with Edit / Pause / Duplicate actions.

Seller storefront

Each seller receives a buyer-facing storefront with cover/profile assets, verified identity signal, rating, completed-order count, campus, description, active categories and tabs such as **All | category filters | Specials**. The visual hierarchy is therefore: **Discovery Card** → **Listing Detail** → **Seller Storefront**.

Asset handling

Listings support a cover image plus gallery. Vossie should validate file type/size, generate optimized thumbnails and responsive derivatives, preserve an appropriate original, and serve media from controlled storage/CDN infrastructure. Category-specific moderation rules apply. Buyer-facing imagery must not intentionally misrepresent the item being sold.

Discovery behaviour

The buyer sees simple top-level category tiles and progressive subcategory/filter chips. Structured search maps natural queries into catalogue attributes—for example, “black hoodie medium” → Fashion / Clothing / Hoodie / Black / M; “Network+ tutor tomorrow” → Services / Tutoring / Networking / date availability; “calculator for three days” → Rentals / Academic / Calculator / rental dates.

Locked principle

One shared presentation system, category-aware transaction logic. Every listing looks recognisably Vossie, while Food, products, Fashion, Services, Rentals and Digital expose only the controls needed for that transaction type.

24. Vossie Reels — Shoppable Video Discovery — Confirmed

Vossie Reels is a first-class Marketplace destination: short-form vertical video tied directly to commerce objects. It is a discovery and conversion surface, not a general social network.

Marketplace navigation

Mobile bottom navigation: Home | Search | Reels | Orders | Me. Reels occupies the centre discovery position. Home also contains a compact “Trending on your campus” Reel preview rail; See All opens the full feed.

VOSSIE REELS	INTERACTION
For You Campus Following	Vertical swipe Previous / next Reel.
FULL-SCREEN SELLER VIDEO 5–30 seconds	Tap seller Open Seller Storefront.
■ Save ■ Share ■ Report	Tap listing Open Listing Detail.
@ThatosKitchen ✓ Fresh Alfredo available today	Multiple tagged listings Horizontal swipe inside the product rail.
SHOPPABLE RAIL [img] Chicken Alfredo R55 · ★4.8 · 7 available VIEW / ADD →	Category-aware CTA Food: Add/Order Product: Add to Cart Service: Book Rental: Check Availability Digital: View/Buy
Home Search REELS Orders Me	Commerce rule Published Reels must remain connected to valid Vossie commerce content.

Seller creation flow

Seller Mode → + → **Create Reel** → **Record or Upload** → **Trim (5–30 sec)** → **Cover Frame** → **Caption** → **Link one or more valid listings** → **Preview** → **Publish**. A seller may also start from a listing via Promote → Create Reel, with that listing pre-linked.

Seller integration

Seller storefront navigation becomes **Shop | Reels | Reviews**. Seller Mode includes Reels alongside operational areas. Seller Studio gains Content/Reels analytics: views, unique viewers, listing taps, store visits, saves, add-to-cart, bookings, purchases and conversion rate.

Discovery and ranking

Feed tabs: **For You | Campus | Following**. Campus is a major pilot signal. The first implementation should use explainable ranking rather than a TikTok-scale recommendation engine: campus relevance, category interest, availability, content/listing quality, freshness, engagement and seller reliability, with mechanisms that preserve exposure opportunities for new sellers.

Commerce graph

Reel → Shoppable Listing Rail → Listing Detail → Seller Storefront → Cart / Booking / Rental / My Library. Reels reuses the catalogue, pricing, promotion, availability, fulfilment and payment architecture already defined; it does not create parallel inventory.

Moderation & media pipeline

Upload processing: **file validation → controlled storage → transcode/optimize → thumbnail/cover → content moderation → linked-listing validation → publish**. Users can report a Reel. Prohibited or misleading content, prohibited goods/services, stolen content, harassment/sexual content and off-platform payment redirection are subject to enforcement. Reel moderation and listing moderation remain separate objects/actions.

Pilot social scope

Do not launch public comment threads in the pilot. Saves, shares and seller/listing navigation provide sufficient commerce interaction without creating a large additional moderation surface. Seller Following can be activated as the follower/notification system matures.

Core data object

MarketplaceReel: reel_id, seller_id, video_asset_id, cover_asset_id, caption, linked_listing_ids, campus_scope, moderation_status, publication_status, created_at. Analytics are stored as structured events rather than mutable vanity counters being treated as transaction truth.

Locked principle

Reels is commerce content. It is a permanent Marketplace navigation destination and a seller discovery tool, but it must continuously lead back into Vossie listings, storefronts and transactions rather than evolving into a generic student social feed.

25. Communications, Notifications & Transaction Documents — Confirmed

Vossie uses one event-driven communications system. Transaction/order/payment/ledger records remain authoritative; the communications layer determines who must be informed, through which channel, using which versioned template.

Core architecture

Domain Event → **Transactional Outbox** → **Notification Orchestrator** → **Channel Workers** → **In-App / Push / Email** → **Delivery Status**. Communications are asynchronous and must not block or roll back a successful commerce transaction.

Buyer Notification Centre

All | Orders | Payments | Messages | Marketplace | Account. Cards show meaningful title, contextual detail, timestamp/unread state and deep-link directly to the underlying order, payment, conversation, listing, dispute or account action.

Channel policy

Event	In-app	Push	Email
Order placed	Yes	Yes	Confirmation
Payment confirmed	Yes	Yes	Receipt/confirmation
Seller accepted / preparing	Yes	Accepted: Yes; preparing: optional	No
Ready / courier assigned / collected / nearby	Yes	Yes	Normally no
Delivered	Yes	Yes	Delivery confirmation
Protection ending / completed	Yes	Protection: Yes	Final record remains available
Refund / dispute milestone	Yes	Yes	Important financial/case milestones
Payout sent	Seller/Courier	Yes	Yes
Security event	Yes	When appropriate	When appropriate

Preference rule

Required transactional, payment, fulfilment and security communications are separated from optional Marketplace/promotional communications. Optional seller/Reels/marketing activity can be disabled without suppressing critical order or account information.

Buyer email/document patterns

Transaction type	Key content	Primary action
Physical / Food	Order reference, seller, items, totals, payment state, fulfilment method, ETA/status; delivery confirmation and protection state after handoff.	Track / View Order
Service	Service, provider, booking date/time, duration, online/in-person location, amount; scheduled reminders.	View Booking
Rental	Item, rental period, rental charge, deposit, collection/return details; due and return/deposit status.	View Rental
Digital	Purchased resource, price, entitlement confirmation. Do not attach seller content directly to email.	Open in My Library
Refund	Order reference, refund amount, processing state, destination/payment method and provider timing caveat.	View Refund

Permanent receipt & documents

Every order exposes **Receipt & Documents**: View Receipt, Download PDF, Email Receipt and Share Receipt. Additional records appear when relevant: payment record, refund record, dispute outcome and rental return record. Receipt HTML and PDF are generated from the same normalized server-side receipt data.

Receipts preserve transaction history. Later refunds/adjustments create linked records and reversing ledger entries rather than silently rewriting the original transaction. Category-specific receipt sections show the relevant fulfilment, booking, rental or digital-entitlement facts.

Seller communications & statements

Seller notifications cover new order/request, payment-backed secured earnings, fulfilment actions, protection/completion, earnings becoming Available and payout. Seller statements expose gross sale, promotion/discount allocation, Vossie seller commission, adjustments/refunds, net seller allocation and lifecycle state: **Secured** → **Available** → **Payout Processing** → **Paid Out**.

Courier communications & statements

Courier notifications cover assignment/offer, pickup/custody actions, delivery exceptions, completion and earnings. Earnings records show gross delivery earning, Vossie courier fee, adjustments if applicable and net courier earning. Values always originate from the ledger.

Operations: Communications Monitor

Operations receives a delivery-monitoring view rather than an unrestricted message-content viewer.

Event	Recipient	Channel	Template	State
ORDER_DELIVERED	Buyer	Push	buyer_order_delivered_v1	Delivered / Failed / Retry
REFUND_COMPLETED	Buyer	Email	buyer_refund_completed_v1	Delivered / Failed / Retry
PAYOUT_SENT	Seller	Email	seller_payout_sent_v1	Delivered / Failed / Retry

Versioned templates

Examples: **buyer_order_confirmed_v1**, **buyer_order_delivered_v1**, **buyer_refund_completed_v1**, **seller_new_order_v1**, **seller_earnings_available_v1**, **courier_delivery_assigned_v1**, **courier_earning_available_v1**, **security_new_login_v1**.

Template revisions do not alter the historical meaning of already-sent communications.

Demo implementation scope

For the working demo, the complete architecture does not require production email/SMS/push infrastructure. Implement the event stream, Notification Centre, order timeline, receipt/document screens and realistic branded email previews with sandbox/mock delivery states. This demonstrates the full workflow without falsely implying live production messaging or payment-provider settlement.

Locked principle

One event. One source of transaction truth. Multiple appropriate communication channels. The commerce domain records what happened; communications communicate that fact without becoming the transaction authority.

26. Integration Architecture & Prototype Simulation Strategy - Confirmed

Vossie separates integration contracts from provider implementations. The demo must visibly distinguish working internal logic, sandbox/provider-backed functions and simulated external behaviour. If Eduvos approves production deployment, provider adapters can be replaced or hardened without redesigning Buyer, Seller, Courier or Operations workflows.

Current research basis

Current provider documentation supports the proposed architecture: Microsoft Entra supports OIDC/SSO and Microsoft Authenticator MFA; Expo supports foreground/background location subject to OS permissions and constraints, and push through Expo or native FCM/APNs; Google Maps Routes supports route calculation/matrices and Geocoding maps addresses to coordinates; Supabase Realtime supports broadcast/presence/database-change delivery; Paystack documents signed-event/webhook workflows and Transfers availability in South Africa. Provider selection remains replaceable through adapters.

APP 1 - Buyer integrations

Integration	Prototype	Production path
Identity	Clearly labelled Eduvos/Microsoft sandbox sign-in + simulated Authenticator approval.	Eduvos-approved Entra/OIDC; institutional MFA remains under Eduvos/Microsoft control.
Maps/location	Real map component, approved campus/hub pins, controlled routes, simulated courier marker.	Routing/geocoding provider + privacy-scoped live courier location for active orders.
Payments	Sandbox checkout + functional Vossie ledger; no real settlement.	Approved SA PSP via Payment Adapter, verified server-side webhooks, reconciliation/refunds.
Realtime	Order state, chat and simulated tracking synchronized through backend realtime events.	Scaled WebSocket/realtime service with authorization per order/resource.
Notifications	Functional Notification Centre + email/push previews or test push.	APNs/FCM/transactional email with delivery monitoring.
Media	Listing/Reel/profile assets in controlled demo storage.	Private/public object policies, CDN, derivatives, scanning/transcoding/moderation.

APP 1 - Seller integrations

Seller reuses Identity, Payments/Earnings, Notifications, Realtime and Media. Additional integration: **Timetable PDF -> isolated upload/processing -> proposed class blocks -> seller review/correction -> availability engine**. Demo may use controlled sample documents; production requires hardened parsing, malware/file validation, retention controls and explicit seller confirmation.

Fulfilment/location UI should prefer approved campus pickup/handoff points rather than exposing unnecessary private residence data.

APP 1 - Courier integrations

Courier is the strongest map/GPS demonstration. Demo flow: **Offer -> Accept -> Pickup/QR -> Start Delivery -> simulated GPS route -> buyer/hub/ops update together -> arrival -> OTP -> Delivered**. A real map is used while location movement is simulated and labelled as such.

Capability	Prototype	Production
Map & route	Campus/hub/buyer route, ETA and moving marker from controlled route points.	Routes/geocoding provider; traffic-aware routing where justified.
Device location	Optional foreground demo location; deterministic simulated route preferred for judging reliability.	Foreground/background device GPS with explicit permissions, throttling, battery/privacy controls and OS-policy compliance.
Navigation	Embedded route view or deep-link to external maps.	External turn-by-turn navigation or approved Navigation SDK; Vossie need not build its own navigation engine.
Proof	Functional QR and demo OTP against demo order state.	Server-validated QR/custody + secure recipient proof and abuse controls.

APP 2 - Campus Hub integrations

Campus Hub uses maps as an operational visualization of custody and transfer state, not as the source of truth. Screens show incoming parcels, ready local deliveries, outgoing transfers, route/ETA and exceptions. The authoritative workflow remains **scan -> verify -> custody -> manifest -> exact-set reconciliation -> depart -> destination reconciliation**.

Demo inter-campus transfer: origin Hub scans packages into a manifest, locks it, animates the transfer along a predefined campus route, destination Hub scans/reconciles the same manifest, then releases to pickup/local courier. Production can attach actual transfer telemetry/provider tracking to the same transfer object.

APP 2 - Incubation integrations

Incubation consumes aggregated marketplace analytics, not individual GPS. Inputs include searches, views, Reels engagement, orders, category demand, inventory gaps, seller performance and campus. Outputs include Demand Intelligence, Trending Categories, Unmet Searches, Seller Growth and programme interventions. Campus maps, if used, display aggregate business signals only.

APP 2 - Platform Operations integrations

Operations is the integration observability and control surface. It consumes identity status, payment/ledger state, logistics maps, notification delivery, moderation/media state, support cases, security events and analytics subject to role permissions.

A demo **Integration Health** panel should explicitly label each dependency: Connected, Sandbox, Simulated, Degraded or Offline. This makes the prototype credible without pretending simulated services are production integrations.

Integration Gateway

Marketplace/Operations clients -> Vossie API -> Domain Services -> Integration Gateway -> provider adapters. Adapters should cover Identity, Maps/Routing, Payments, Notifications, Media/Storage, Document Processing, Moderation and Analytics/Observability. Domain logic must not depend directly on a specific vendor SDK where an adapter boundary is practical.

Prototype vs production matrix

Area	Demo target	If selected for production
Maps	Real map UI + predefined campus/hub routes	Production routing/geocoding and cost/usage controls
Courier GPS	Simulated deterministic movement	Permissioned device GPS + background strategy
Hub transfer	Real manifest state machine + animated route	Operational transfer/provider telemetry
Eduvos identity	Sandbox institutional flow	Eduvos-approved Entra/OIDC + institutional MFA
Payments	Sandbox PSP + real demo ledger	Approved PSP, webhooks, reconciliation, refunds/payouts
Push/email	In-app + previews/test delivery	APNs/FCM + transactional email provider
Realtime	Functional demo events	Hardened authorized realtime infrastructure
Reels/media	Upload/playback where feasible	CDN, transcoding, scanning, moderation
Timetable	Controlled sample extraction	Hardened isolated document processing
QR/OTP	Functional demo state transitions	Server validation, anti-replay/abuse controls
Analytics	Live demo events + seeded history	Production event pipeline + governed retention

Demo Scenario A - Local order

Buyer order -> sandbox payment -> seller receives order -> seller prepares -> Hub QR intake -> courier offer/accept -> route appears -> simulated courier GPS -> Buyer + Hub + Ops update in sync -> OTP -> Delivered -> protection state -> ledger/earnings transition -> receipt + notifications -> analytics event.

This single scenario proves cross-role synchronization, maps, logistics, payments, communications, QR/OTP, ledger and analytics without requiring production money movement or unrestricted live tracking.

Demo Scenario B - Hub-to-hub

Seller -> origin Campus Hub -> QR intake -> transfer manifest -> exact-set scan -> manifest lock -> animated inter-campus route -> destination Hub -> reconciliation -> local courier/pickup -> OTP -> completion.

This demonstrates that Vossie is a campus commerce network with custody and transfer logic rather than a simple marketplace catalogue.

Implementation safeguards from current research

1. Background GPS is permission- and OS-constrained; do not promise uninterrupted tracking. The demo should use deterministic simulated movement while production uses explicit foreground/background permission flows and handles platform limits.
2. Payment truth comes from server verification/webhooks and idempotent ledger processing, never from the client success screen.
3. Identity uses institutional OIDC/SSO if Eduvos approves it; Vossie does not collect Eduvos passwords or Microsoft Authenticator secrets.
4. Realtime delivery is a transport layer, not authorization. Every subscription/event remains scoped to authenticated resources.
5. Private digital resources should use time-limited authorized access rather than permanent public URLs.

Locked principle

Build the demo around stable Vossie domain contracts and provider adapters. Demonstrate the complete user experience now with sandbox, functional internal state and clearly labelled simulations; replace simulations with approved production integrations later without rebuilding the product model.

27. Courier Eligibility, Campus Transfers & Buyer Requests - Confirmed

This section consolidates the three confirmed marketplace/logistics decisions following the integration architecture in Section 26.

27.1 Courier Registration & Eligibility

Eduvos student verification establishes identity, but does not automatically grant Courier capability. Courier is an optional capability on the same Vossie identity and is approved independently from Buyer or Seller access.

Flow: Verified Eduvos Student -> Activate Courier -> Select one or more transport modes -> Mode-specific verification -> Approval -> Configure availability/radius -> Select today's transport mode -> Receive eligible delivery offers.

Mode	Onboarding/verification	Dispatch implications
Walking	Student identity, courier terms, campus, availability and practical operating radius.	Short-distance/light jobs only, based on configured rules.
Bicycle	Student identity plus basic bicycle/operational details where justified. Do not invent a driving-licence requirement for an ordinary pedal bicycle.	Distance/size/weight rules appropriate to bicycle delivery.
Motorcycle / Scooter	Appropriate driver and vehicle documentation, expiry/status and operational review.	Only offered jobs compatible with approved mode and capability.
Car	Appropriate driver's licence, vehicle details/supporting documents, expiry/status and operational review.	Eligible for larger/longer local jobs subject to configured limits.

Each transport mode has its own verification state. A document expiry or failed reverification disables only the affected courier capability; it does not remove Buyer/Seller access or unrelated approved courier modes.

When going online, a multi-mode courier selects the transport method being used for that session. Dispatch intersects **campus + current mode + verification + availability + distance + parcel weight/size + fulfilment requirements**. Limits remain configurable by Operations rather than being permanently hard-coded.

Courier offers use an Uber-like card: map, pickup, destination, distance/ETA, package class, expected earning, countdown where genuinely time-sensitive, Accept and Decline. Verification status and courier reputation remain separate concepts.

Prototype: verification is explicitly marked demo/sandbox verification. **Production:** approved document/identity validation processes replace the simulation; Vossie must not assume direct government database access until such access is confirmed.

Operations adds Courier Management: Applications, Approved, Pending, Reverification, Suspended and Incidents. Sensitive verification documents use restricted permissions and audit logging.

27.2 Campus Transfer Network

Cross-campus fulfilment is a hub network, not a default long-distance trip by one student courier. Routing chooses the **nearest viable onward hub**, considering direction toward destination, active routes, schedules, parcel compatibility, capacity, cost and expected arrival - not geographic proximity alone.

Buyer fulfilment choice: where viable, checkout can present Vossie Campus Transfer, an integrated third-party courier option, and other eligible collection/fulfilment methods. The buyer sees price/ETA information appropriate to each option.

A single shipment may contain multiple transfer legs. Each leg has a manifest and custody evidence. At every hub: arrival scan -> manifest reconciliation -> custody recorded -> next viable leg -> outgoing manifest. Mismatches or missing packages create exceptions rather than silent status changes.

Shipment lifecycle: Seller Ready -> Origin Hub Received -> Waiting for Transfer -> Manifested -> In Transfer -> Intermediate Hub Received -> Next Transfer -> Destination Hub Received -> Local Delivery Ready -> Courier Collected -> Delivered.

The buyer sees one continuous shipment timeline/map with campus milestones. Hub and Operations users see the detailed legs, manifests, custody holders, reconciliations and exceptions.

The production Campus Route Registry records origin, next hub/destination, distance, transfer method, operating days, cut-off, expected duration, capacity, parcel classes, cost model and Active/Suspended state. Routing can select another permitted path when a route is unavailable.

Vossie should not force inefficient multi-hop routing. Where the campus route is unreasonable, unavailable or materially inferior, an integrated third-party shipment can be offered. Provider events map back into the Vossie timeline while the provider remains authoritative for its own logistics events.

Local Courier and Campus Transfer Courier/Operator are distinct capabilities. Approval for local car delivery does not automatically authorize inter-campus transfer work.

Prototype: selected campus nodes and routes are predefined and movement is simulated, while QR scans, manifests, reconciliation, custody and shipment state transitions are functional.

27.3 Buyer Request Marketplace

Requests sit beside Listings. Search answers 'what supply exists now'; Request answers 'I need this - who can fulfil it?'. A request is a demand-discovery route into the existing transaction engine, not a second commerce engine.

Flow: Request -> category-specific details -> matching -> eligible sellers notified -> structured seller offers -> buyer compares -> accepts one offer -> normal Order/Booking/Rental -> payment -> fulfilment -> receipt/ledger/notifications.

Request creation and matching

Buyer Home/Search includes **Can't find what you need? Request it**. Initial request types are Product, Food, Service, Rental and Digital. The existing taxonomy dynamically controls fields such as quantity, variant, condition, timing, duration, campus, fulfilment, budget, dietary/allergen information, rental period and optional media.

Before broad matching, Vossie may surface compatible existing listings so buyers can transact immediately. If the buyer continues, matching uses category/subcategory, campus, seller capability, availability, inventory/service type, fulfilment radius and seller status. Requests are not indiscriminately broadcast to every seller.

Seller offers and buyer decision

Eligible sellers receive an Uber-like request card and may Respond or Dismiss. Responding creates a structured offer containing price, timing/availability, relevant fulfilment or service terms and an optional concise note. Sellers do not see competing sellers' offer prices, avoiding an uncontrolled visible bidding race.

The buyer receives real-time 'New offer' notifications and an Offers view containing the information needed to choose. Accepting an offer locks/converts it into the existing transaction workflow; payment, ledger, protection, fulfilment, receipts and communications are reused.

Request lifecycle and controls

Draft -> Open -> Matching -> Offers Received -> Offer Accepted -> Converted to Transaction -> Fulfilled -> Closed. Alternate states include Expired, Cancelled, No Match and Moderation Hold. My Requests provides Active and Closed views; Seller Mode adds For You, Responded, Won and Closed.

Requests cannot bypass marketplace controls. Food requests only reach sellers eligible for the relevant food category/compliance state. Digital/academic requests remain subject to rights, academic-integrity and prohibited-content rules.

Incubation and Operations value

Unfulfilled and partially fulfilled requests become structured demand intelligence: category, campus, request volume, offer rate and fulfilment rate. Incubation uses this to identify unmet demand and seller opportunities. Operations receives request moderation/abuse tooling without manually controlling normal matching.

Prototype proof flow

Buyer requests a Network+ tutor tonight -> matching identifies eligible sellers -> Seller A sends a structured offer -> Buyer receives the offer in real time -> Buyer accepts -> Vossie creates the booking -> sandbox payment -> seller confirmation -> notifications/receipt/ledger update -> analytics records the conversion.

Locked combined principle

Courier capability is verified and mode-aware; cross-campus logistics is a custody-controlled hub network with optional third-party fulfilment; and buyer Requests expose unmet demand while converting successful matches into the same trusted Vossie transaction system.